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Training School on Mathematical Oncology

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Mini-Project: *Developing a Machine Learning Pipeline to Characterise Tumour Core and Periphery on Glioblastoma Spatial Data*

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Motivation

Glioblastoma (GBM) is the most aggressive diffuse primary brain tumour in adults. Its poor outcome is linked not only to rapid growth but also to extensive infiltration: malignant cells disperse beyond the tissue that can be identified and removed surgically, remain within the surrounding brain and contribute to near-inevitable recurrence. A specimen described as "tumour core" therefore coexists with a spatially extended periphery containing variable mixtures of infiltrating malignant cells, neurons, oligodendrocytes, astrocytes, microglia, vascular cells and extracellular tissue components [**1,2**].

This spatial heterogeneity matters biologically. Local nutrient and oxygen availability, vascular remodelling, immune-cell composition and interactions with normal neural lineages can favour different malignant states and treatment responses. Bulk sequencing averages these signals over an entire specimen. Single-cell RNA sequencing resolves cell populations but normally loses their original tissue coordinates. Spatial transcriptomics addresses this gap by measuring gene expression at thousands of positions while preserving where each measurement was made.

The human GBM atlas developed by Sonpatki and colleagues combines spatial gene-expression profiles, high-resolution histology, matched single-cell data and subcellular spatial measurements to describe anatomically organised tumour niches. The atlas reveals a structured transition from leading-edge and infiltrating regions towards cellular tumour, vascular, immune-glial and hypoxic niches, rather than a homogeneous mass [**2,3**]. This creates a natural quantitative problem: can a machine-learning (ML) model recognise an operational distinction between core and periphery in tissue from a patient that was not used to train the model?

This Mini-Project uses two related feature representations at each tissue spot: measured gene expression and metabolic-module activity computationally inferred from that expression. The aim is not merely to obtain a high classification score. It is to determine whether the result generalises across patients, which biological signals drive the prediction, whether the metabolic representation adds useful information and where the classifier fails in tissue space. No prior coding experience is assumed: the project is supported by an easy-to-follow Python notebook in which the principal technical steps are demonstrated and then reproduced or adapted by each group.

Central question: Which transcriptomic and transcriptome-inferred metabolic programmes distinguish tumour core from tumour periphery in an unseen patient, and how robust and biologically interpretable is that distinction?

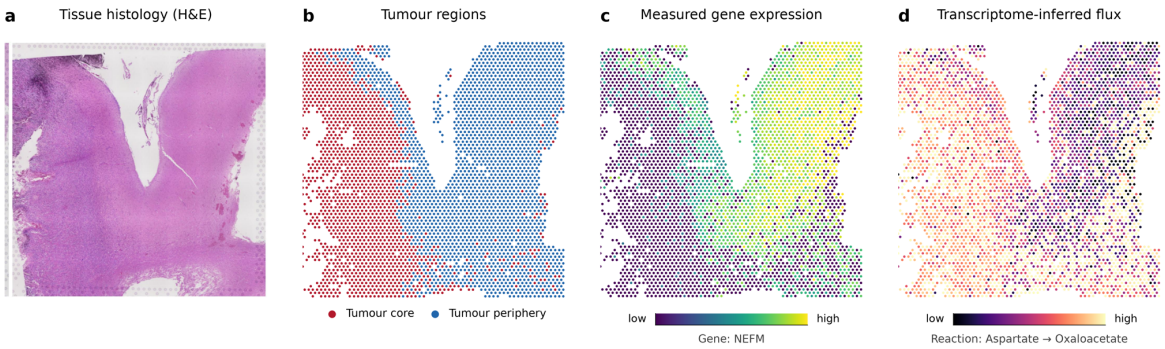


Figure 1: Project data modalities: (a) H&E tissue image; (b) binary regional labels supplied for the project; (c) measured spatial expression of the example gene *NEFM*; and (d) an example transcriptome-inferred metabolic flux score. The metabolic layer is derived computationally from the expression profile of the same spot.

Specific biological context of the problem

A binary task extracted from a richer spatial architecture. The complete atlas distinguishes cellular tumour (CT), infiltrating tumour (IT), leading edge in grey or white matter, immune-glial niches, vascular regions and hypoxic or perinecrotic compartments. The underlying regional framework is based on the Ivy Glioblastoma Atlas Project (Ivy GAP), which defines histologically recognisable regions of GBM tissue, and was refined by Sonpatki and colleagues for the tissue sections in their atlas. The present teaching dataset reduces part of this architecture to the operational classes tumour core and tumour periphery. The exact mapping from the original annotations to these two labels must be recorded in the accompanying data dictionary. "Periphery" must not be interpreted as purely healthy brain, nor "core" as a single homogeneous cell population.

Core and peripheral tissue differ in composition. In matched specimens, core samples contain a larger malignant-cell fraction. Proliferating and mesenchymal-like tumour states, blood-derived macrophages, proliferating microglia and regulatory T cells are more prominent in core tissue, whereas resting microglial populations are more prevalent in peripheral tissue. The atlas also identifies an oligodendrocyte state, Oligo_2_3_2, enriched in core and perivascular regions and associated with adverse clinical features [**2**]. Along the spatial transition from white-matter leading edge to infiltrating tumour and cellular tumour, normal neural-lineage signals progressively coexist with reactive astrocytes, microglia and emerging malignant-cell programmes.

A spot is a tissue mixture, not an individual cell. A Visium spot has a diameter of approximately 55 μm and can contain transcripts from several cells. The classifier therefore predicts a tissue-region label from a mixed molecular profile; it does not identify the phenotype of one cell. Important predictors may reflect malignant-cell programmes, local non-malignant composition or both.

The labels have provenance. In the source atlas, anatomical-feature assignment combines histology, marker-gene scores, inferred cell states and hierarchical rules. Some predictor genes may consequently be related to the rules used to define the labels. A robustness analysis that removes annotation-marker genes is therefore required before interpreting a gene as an independently discovered regional signature.

The metabolic variables are inferred, not directly measured. The 168 metabolic-module values are estimated with scFEA, a graph neural network that infers reaction-level metabolic flux from transcriptomes while constraining its predictions through the stoichiometry of a metabolic module network [**4**]. They may compress correlated expression into a biologically meaningful representation, but they are not independent metabolomic measurements and should be described as transcriptome-inferred activity or flux scores.

Source article, anatomical atlas and scFEA resources

The links for direct access:

- Sonpatki *et al.*, *Nature Communications* (2026): https://www.nature.com/articles/s41467-026-69716-2
- Ivy Glioblastoma Atlas Project: https://glioblastoma.alleninstitute.org/
- scFEA method paper: https://genome.cshlp.org/content/31/10/1867
- scFEA software repository: https://github.com/changwn/scFEA

Dataset and machine-learning formulation

The project subset contains IDH-wild-type GBM tissue from six patients and 15 sections (2–5 sections per patient), profiled with the 10x Genomics Visium platform, with approximately 30,000 spatial spots. It is derived from the spatial atlas of Sonpatki *et al.* **[2]**. Every spot has a patient identifier, section identifier, spatial coordinates, a binary regional label, a normalised gene-expression profile and 168 transcriptome-inferred metabolic-module scores.

Data component	Scale	Dimension	Role in the project
Metadata	Spot	6 patients; 15 sections	Patient, 2–5 sections per patient, spatial coordinates, preservation or batch variables when available, and core/periphery label
Gene expression	Spot	27,573 genes	Normalised transcript counts measured with 10x Genomics Visium, together with the spot locations; a filtered subset will normally be used for the core analysis
Metabolic representation	Spot	168 modules	Activity of each metabolic module, estimated computationally from the expression profile of the same spot using scFEA

The files that the group will use will be pre-aligned by spot identifier and provided in ready-to-load tabular and sparse-matrix form, so that no afternoon is lost to raw-data wrangling. A data dictionary will document the anatomical-label mapping, coordinates, feature names and available technical covariates.

For spot i, let $y_i = 1$ denote core and $y_i = 0$ denote periphery. Let $\mathbf{x}_i$ contain the selected gene features, metabolic features or their combination. The core classifier is a regularised logistic regression,

$$p_i = \Pr\left(y_i = 1 \mid \mathbf{x}_i\right) = \frac{1}{1 + \exp\left[-\left(\beta_0 + \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\beta}\right)\right]}.$$

An elastic-net fit estimates the coefficients by minimising the training loss plus a mixture of L^1 and L^2 penalties,

$$\mathcal{J}(\beta_0, \boldsymbol{\beta}) = -\frac{1}{n_{\mathrm{tr}}} \sum_{i \in \mathcal{I}_{\mathrm{tr}}} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i)\right] + \lambda \left[\alpha \|\boldsymbol{\beta}\|_1 + \frac{1 - \alpha}{2} \|\boldsymbol{\beta}\|_2^2\right].$$

Here λ controls the overall regularisation and α balances sparsity against coefficient shrinkage **[5]**. The principal comparison uses three feature configurations: genes only, metabolic modules only, and genes plus modules.

Non-negotiable validation rule

All spots and all sections from the same patient must remain in the same fold. Scaling, missing-value handling, gene filtering, feature selection and hyperparameter tuning must be learned from training patients only. A random split of neighbouring spots would leak patient- and section-specific information and would not test generalisation to a new patient.

Recommended baseline hierarchy

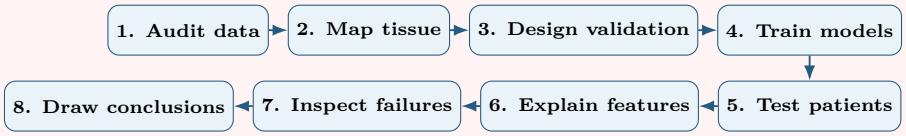
Compare: (i) a majority-class predictor; (ii) metabolic modules only; (iii) a reduced gene-expression model; and (iv) the combined representation. Use patient-grouped cross-validation for model selection and reserve at least one patient, or one full outer fold, for the final untouched evaluation. Class weights may be used when the labels are substantially imbalanced.

Teaching support and Python protocol

No prior coding experience is required. An easy-to-follow, step-by-step Python notebook will supply the data-loading, sparse-matrix, plotting and model-fitting scaffolding. Short demonstrations will introduce each technical step; students will then repeat or adapt it and remain responsible for the validation design, feature comparison, interpretation and final scientific claims.

Questions to address

The questions form one connected pipeline. Decisions made in data auditing and validation must be fixed before interpreting model performance or biological features.



Q1. What exactly is contained in the teaching dataset? Verify patient and section counts, class frequencies, missing values, duplicated spots, coordinate ranges and available technical covariates. Produce a patient-by-class table and determine whether every patient contributes both classes. State the precise mapping used to create the binary core/periphery label.

Q2. How are the labels and molecular signals organised in tissue space? For at least two representative sections, plot the supplied labels, one biologically interpretable gene, one metabolic module and simple spot-level quality measures. Identify whether apparent class differences are global, patient-specific, boundary-localised or associated with a technical variable.

Q3. What validation design answers the scientific question? Construct grouped folds at the patient level. Draw a diagram showing which patients enter training, hyperparameter selection and final testing. Implement all preprocessing within the corresponding training fold. As a pedagogical comparison, assess whether and by how much a naive random spot split overestimates performance.

Q4. Which parsimonious models should be compared? Fit the baseline hierarchy and tune the elastic-net parameters using grouped inner validation. Explain how genes were filtered or selected, how class imbalance was handled and why raw section coordinates were not used as ordinary predictors. Record runtime and the number of non-zero coefficients.

Checkpoint at the end of Wednesday

Before examining feature importance, every group should have: a documented binary label, an auditable patient-level split, one majority baseline, three regularised logistic-regression models and a table of held-out-patient predictions. High apparent accuracy under a random spot split is not an acceptable endpoint.

Questions to address – continued

Q5. Does the classifier generalise across patients? Report a confusion matrix, balanced accuracy, sensitivity, specificity, macro-F1, area under the ROC curve and, when class imbalance is relevant, area under the precision–recall curve. Assess probability calibration with the Brier score or a calibration plot. Give results for each held-out patient where the metric is defined, together with pooled and macro-averaged summaries; one large section must not dominate the conclusion.

Q6. Which genes and metabolic modules contribute reproducibly? For the linear models, examine coefficient sign and magnitude after standardisation. Complement this with held-out permutation importance. Report how often each feature is selected across patient-grouped folds and whether its direction is consistent across patients. Distinguish a stable predictive association from a causal mechanism or therapeutic target.

Q7. Where, and why, does the model fail? Map predicted core probability, uncertainty and misclassified spots back onto the tissue. Test whether errors cluster near core–periphery boundaries or in particular sections. Determine whether important genes mainly reflect malignant-cell programmes, normal neural-lineage composition or immune and vascular components. Repeat the analysis after excluding genes used directly in the original annotation rules and interpret any performance loss.

Q8. What can be concluded from the comparison of data representations? Decide whether the metabolic-module model preserves, improves or reduces patient-level performance relative to gene expression, and whether the combined model adds information after accounting for regularisation. Formulate one biological interpretation supported by the data, one limitation that prevents a stronger claim and one additional dataset or experiment that would provide a more independent validation.

Suggested three-afternoon progression

Tuesday: biological introduction, data dictionary, cohort audit, spatial visualisation and baseline model. **Wednesday:** patient-grouped pipeline, feature filtering, elastic-net tuning and comparison of gene-only, module-only and combined models. **Thursday:** held-out evaluation, feature stability, marker-exclusion analysis, spatial error maps and presentation preparation. **Friday:** 6–8 minute group presentation and discussion.

Interpretation boundary

Good discrimination does not prove that the selected features cause invasion or recurrence. The task evaluates prediction of supplied spatial annotations. Its strongest result is a reproducible, patient-generalizable molecular distinction whose dependence on annotation markers, tissue composition and technical variation has been examined transparently.

Expected outputs, group organisation and optional extensions

Required outputs. Each group should produce: (i) a cohort and class-distribution table; (ii) spatial maps of labels and representative molecular features; (iii) a diagram of the patient-grouped validation design; (iv) a table comparing the baseline, module-only, gene-only and combined models; (v) a stable-feature summary across folds; (vi) a spatial prediction or error map; and (vii) a 6–8 minute Friday presentation in which all group members contribute. The presentation must explicitly cover the question addressed, the main findings from exploration of the gene and metabolic datasets, the ML model used to classify spots as core or periphery, and the genes and metabolic modules judged most relevant through an explainability analysis.

Suggested division of labour. One trainee can lead data auditing and spatial visualisation, one can lead model implementation and grouped validation, and one can lead feature interpretation and presentation design. Analytical decisions and final conclusions should be agreed jointly.

Optional advanced directions. Compare the linear model with a tree-based classifier; formulate a three- or four-class problem using cellular tumour, infiltrating tumour and leading-edge regions; use spatially blocked validation within an unseen section; add neighbourhood-averaged features or graph regularisation; aggregate spots into patient- or section-level pseudobulk profiles; or test the workflow on an independent spatial GBM cohort.

Participant package. The recommended package contains this handout, a sparse gene-expression object, a metabolic-module matrix, a spot-metadata table, section images and a data dictionary defining the binary label and annotation-marker genes. These files will already be organised and aligned, so students can begin analysis without preliminary data wrangling. The easy-to-follow, step-by-step Python notebook should demonstrate data loading, exploration and the core regularised classifier, while leaving validation, model comparison and interpretation decisions to the group.

Key bibliography

[1] Weller M, Wen PY, Chang SM, et al. *Glioma.* Nature Reviews Disease Primers **10**, 33 (2024).

[2] Sonpatki P, Park HJ, Xing YL, et al. *A spatially resolved human glioblastoma atlas reveals distinct cellular and molecular patterns of anatomical niches.* Nature Communications **17**, 2951 (2026). https://www.nature.com/articles/s41467-026-69716-2.

[3] Puchalski RB, Shah N, Miller J, et al. *An anatomic transcriptional atlas of human glioblastoma.* Science **360**, 660-663 (2018).

[4] Alghamdi N, Chang W, Dang P, et al. *A graph neural network model to estimate cell-wise metabolic flux using single-cell RNA-seq data.* Genome Research **31**, 1867-1884 (2021).
Paper: https://genome.cshlp.org/content/31/10/1867;
Code: https://github.com/changwn/scFEA.

[5] Zou H, Hastie T. *Regularization and variable selection via the elastic net.* Journal of the Royal Statistical Society: Series B **67**, 301-320 (2005).

[6] Singhal A, Zhao X, Wall P, et al. *The hallmarks of predictive oncology.* Cancer Discovery **15**, 271-285 (2025).

Scope and reporting

This Mini-Project is a proof of concept based on a restricted research dataset. It does not produce a clinical diagnostic tool, define a surgical margin or establish independent metabolic flux measurements. Reproducibility requires reporting patient splits, preprocessing choices, software versions, random seeds, hyperparameters and the exact feature lists used in every analysis.